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A Central University

DSC-503: COMPUTER NETWORKS

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UNIT 1 : INTRODUCTION

- Introduction to Computer Networks,
- LAN,
- MAN,
- WAN,
- Uses of Computer Networks,
- LAN Technologies- Transmission Topologies, Access methods.
- Network Architecture, Protocol and standards,
- References Model: OSI-ISO and TCP/IP,
- Overview: Circuit switching, Message switching and Packet switching.

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PROTOCOL LAYERING

- In data communication and networking,
  - a protocol defines the **rules** that both the **sender and receiver** and all **intermediate devices** need to follow to be able to **communicate effectively**.

- communication is simple: one simple protocol.
- communication is complex: divide the task between different layers.
  - protocol at each layer, or protocol layering.

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Scenario 1

- Simple Communication between two neighbors.

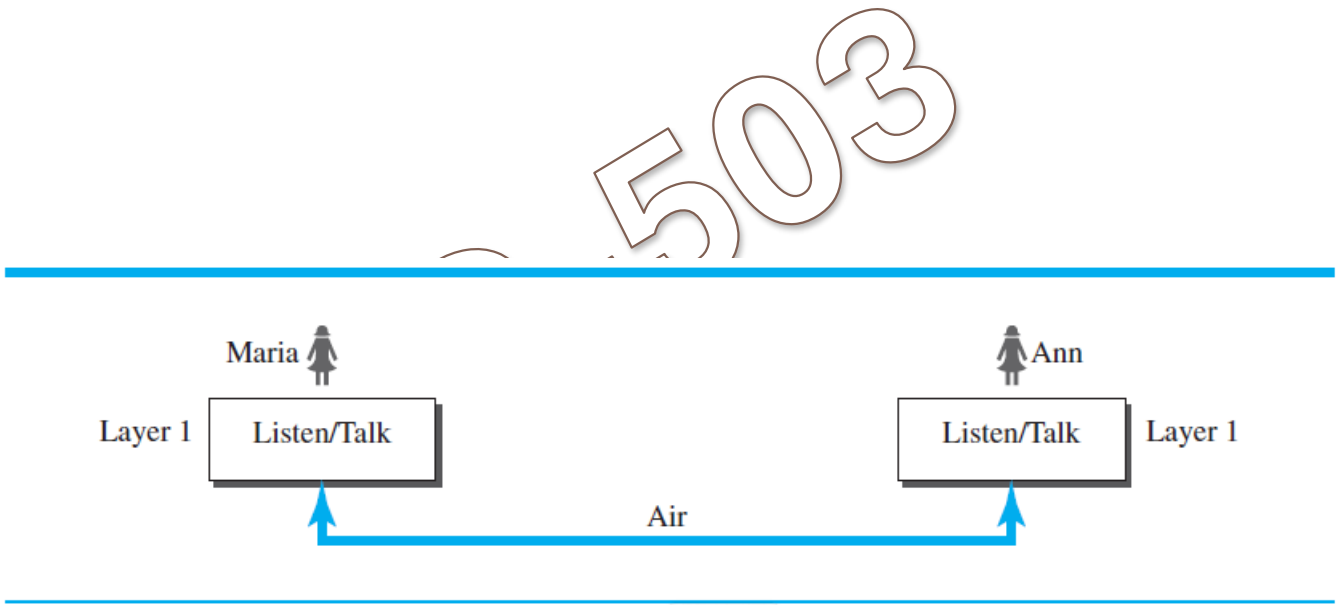


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Scenario 2

- Complex Communication between two People.

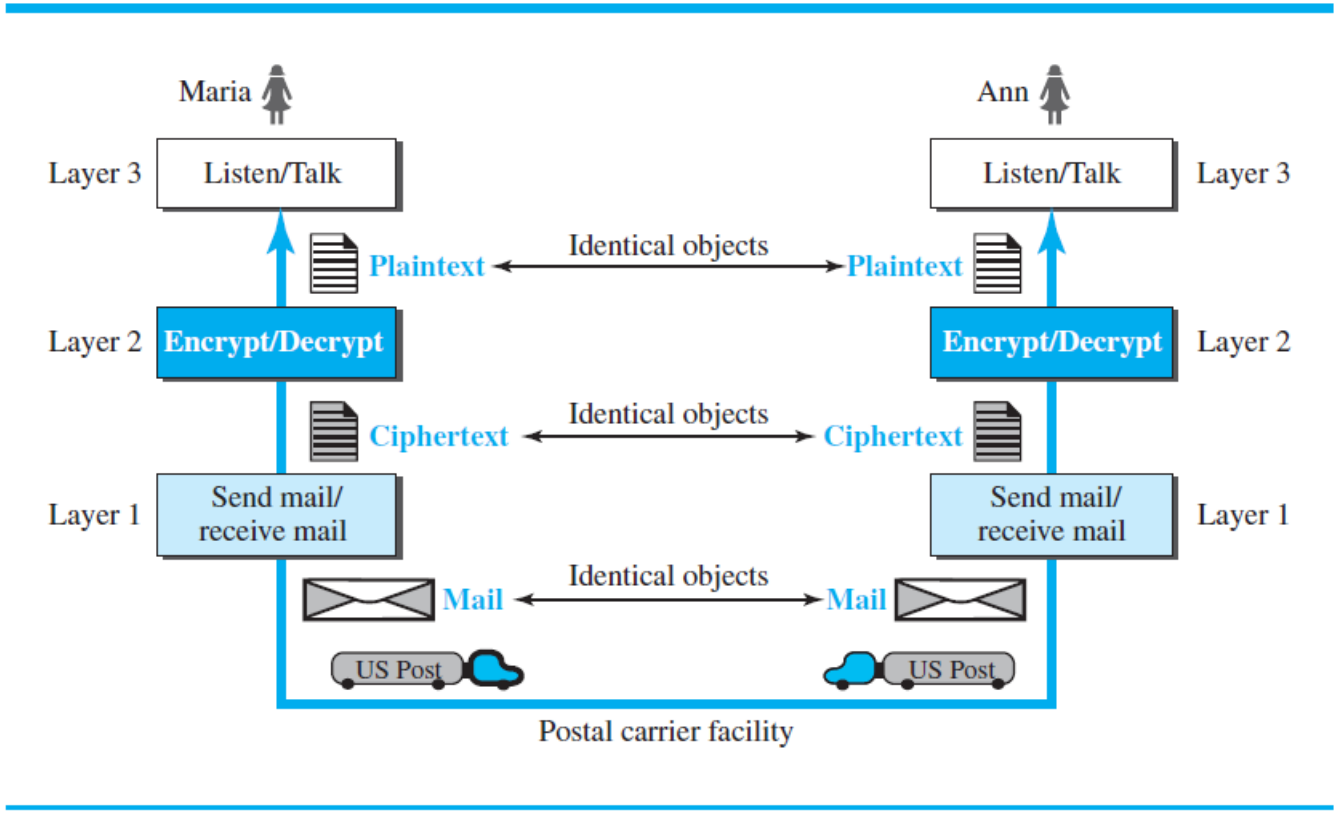


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Principles of Protocol Layering and Logical Connection

- The first principle dictates that if we want bidirectional communication,
  - we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.
- After following the above two principles, we can think about logical connection between each layer.
- This means that we have layer-to-layer communication.
- Two people can think that there is a logical (imaginary) connection at each layer through which they can send the object created from that layer.

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Logical Connection



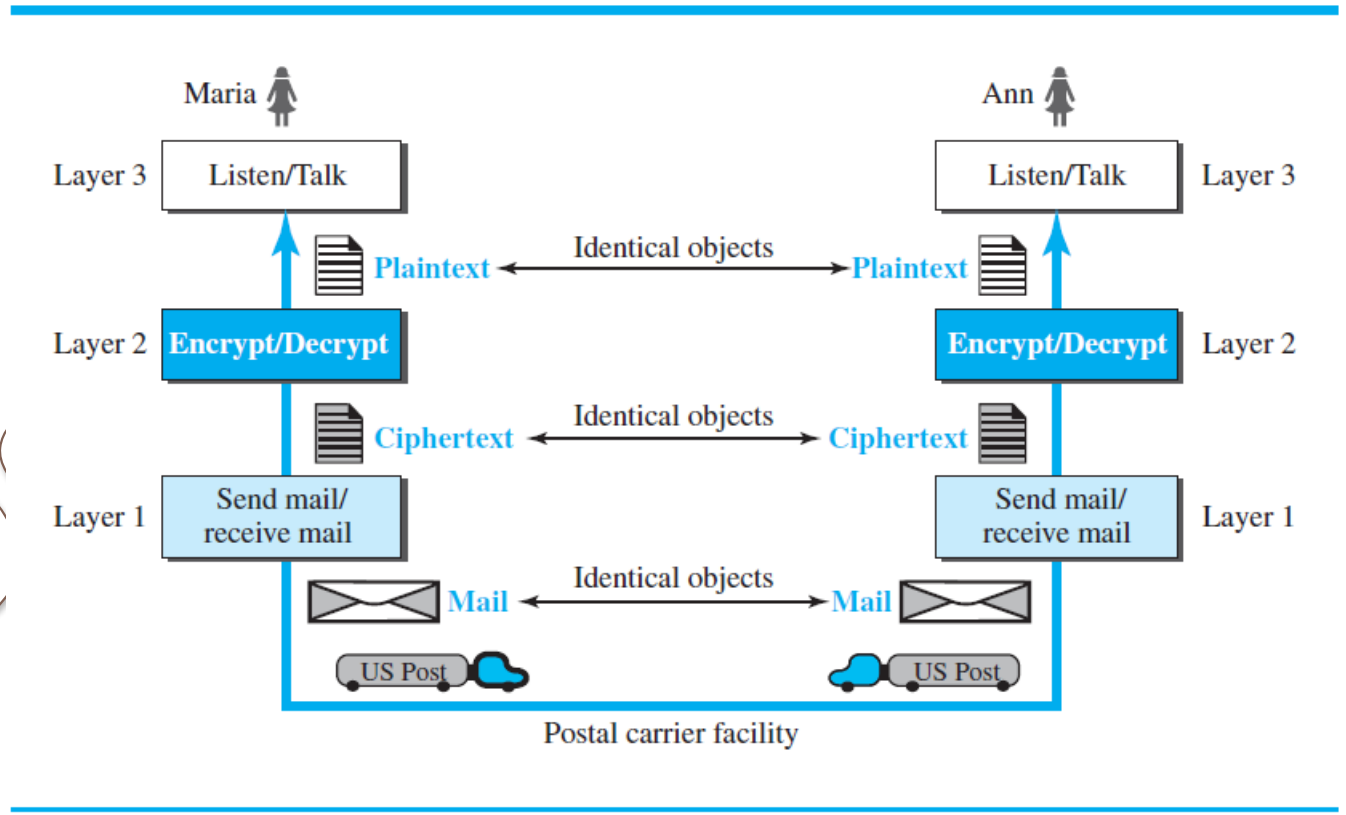


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

THE OSI MODEL

- Open Systems Interconnection introduced in the late 1970s.
- An **open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- It facilitate **communication between different systems** without requiring changes to the logic of the underlying hardware and software.
- It is a model for understanding and designing a network architecture that is **flexible, robust, and interoperable**.

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THE OSI MODEL: 7 Layers

- It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.

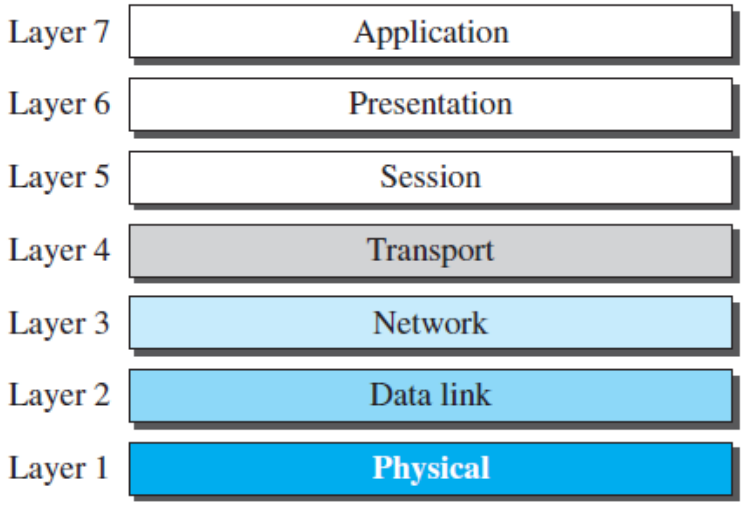


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Physical layer

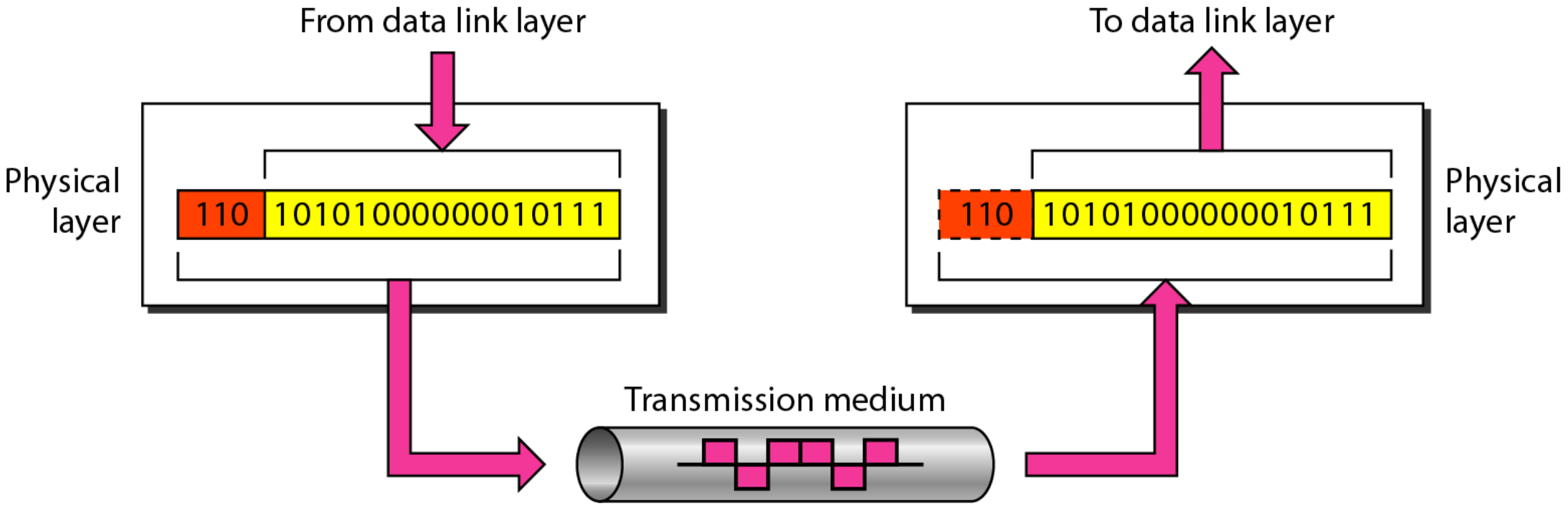


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

The physical layer is also concerned with the following:

- Physical Characteristics of interfaces and medium
- Representation of bits
- Data rate
- Synchronization of bits
- Line configuration
- Physical topology
- Transmission mode

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Data Link layer

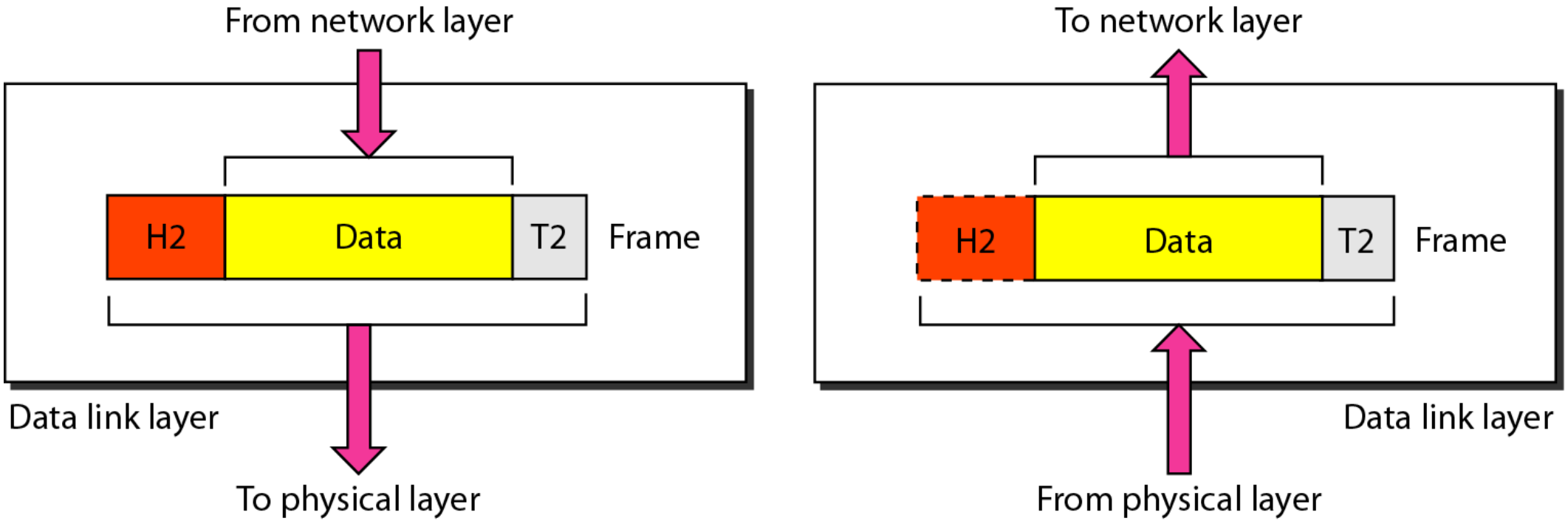


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Responsibilities of Data link includes the following:

- Framing
- Physical addressing
- Flow control
- Error control
- Access control
- Hop to Hop Delivery

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Network layer

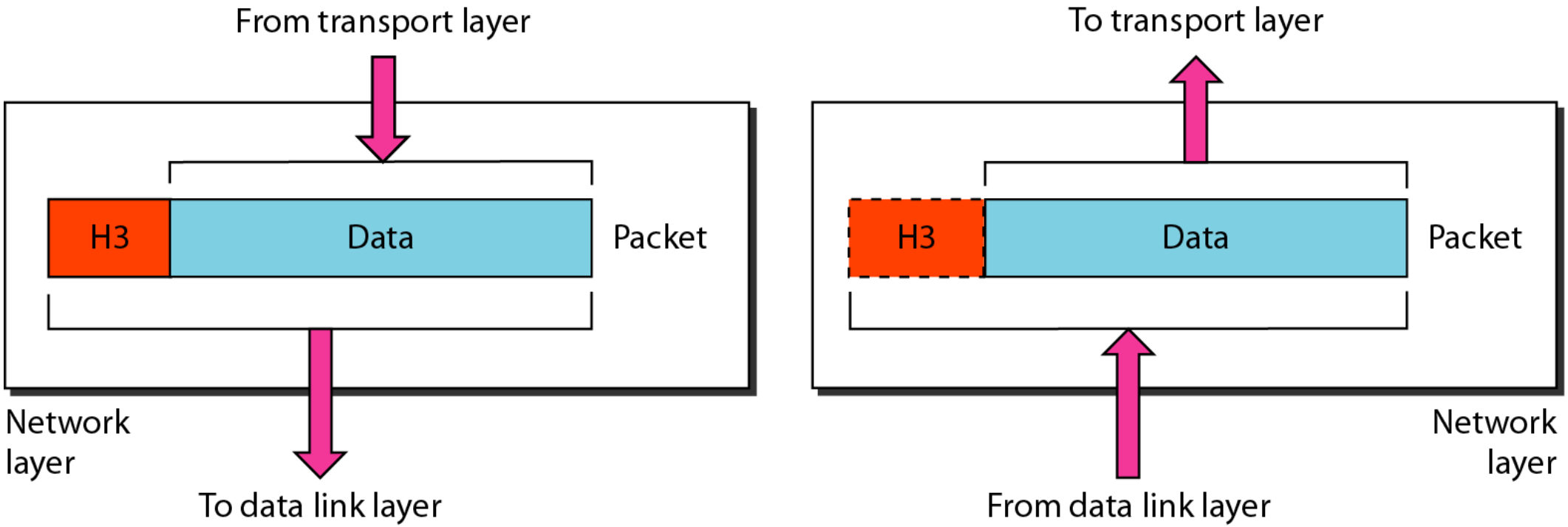


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Responsibilities of Network Layer includes the following:

1. Logical addressing
2. Routing
3. Source to Destination Delivery

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Transport Layer

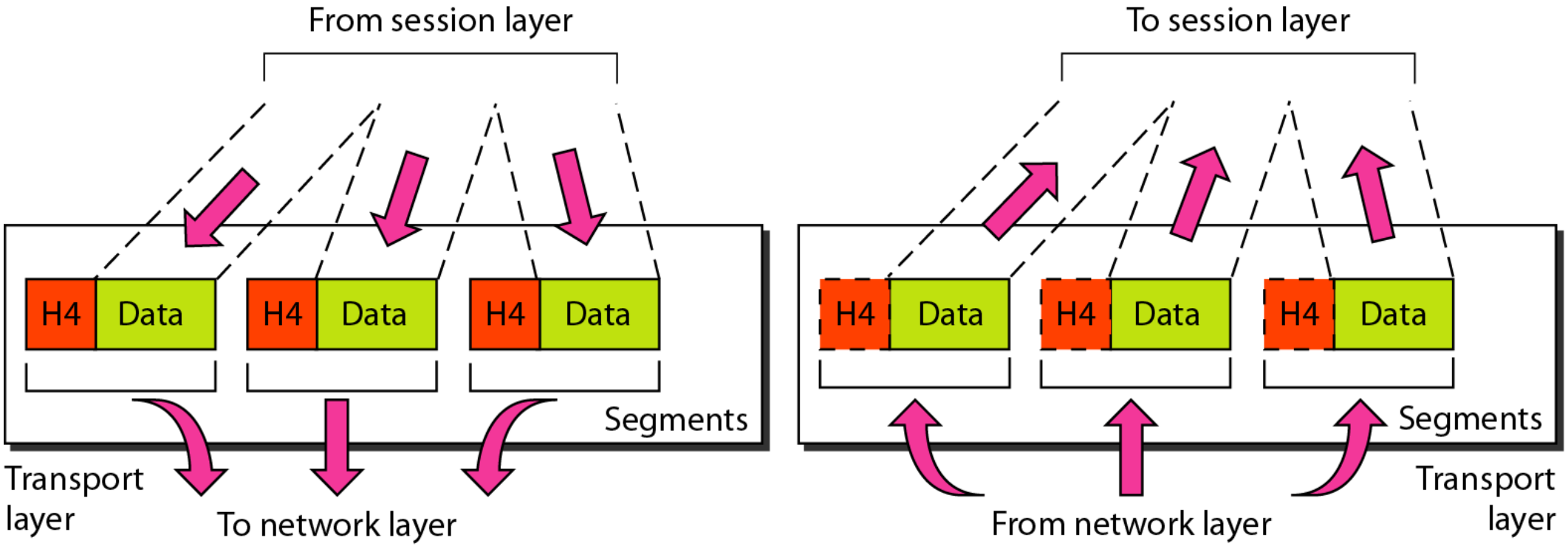


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The transport layer is responsible for the delivery of a message from one process to another.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Responsibilities of Transport Layer includes the following:

1. Service point addressing/ port addressing
2. Segmentation and reassembly
3. Connection control
4. Flow Control
5. Error Control

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Figure *Reliable process-to-process delivery of a message*

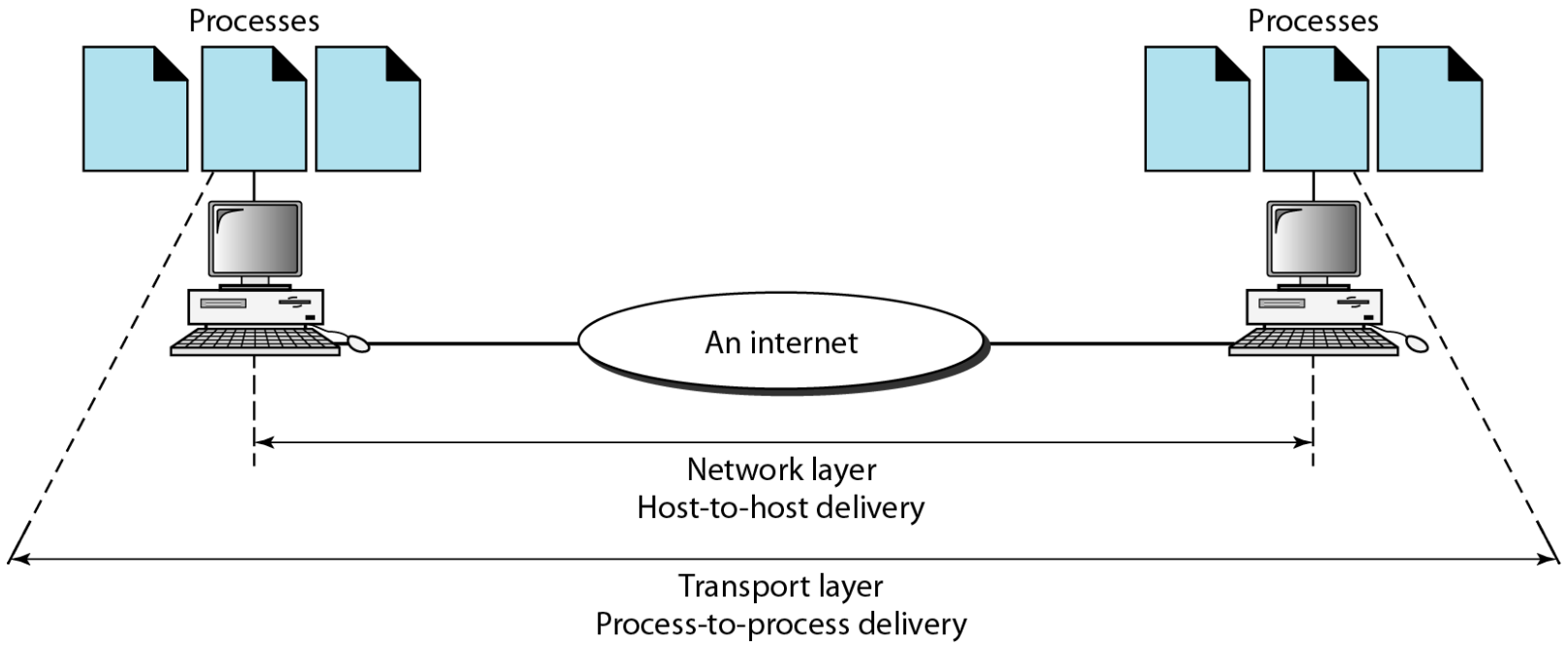


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Session Layer

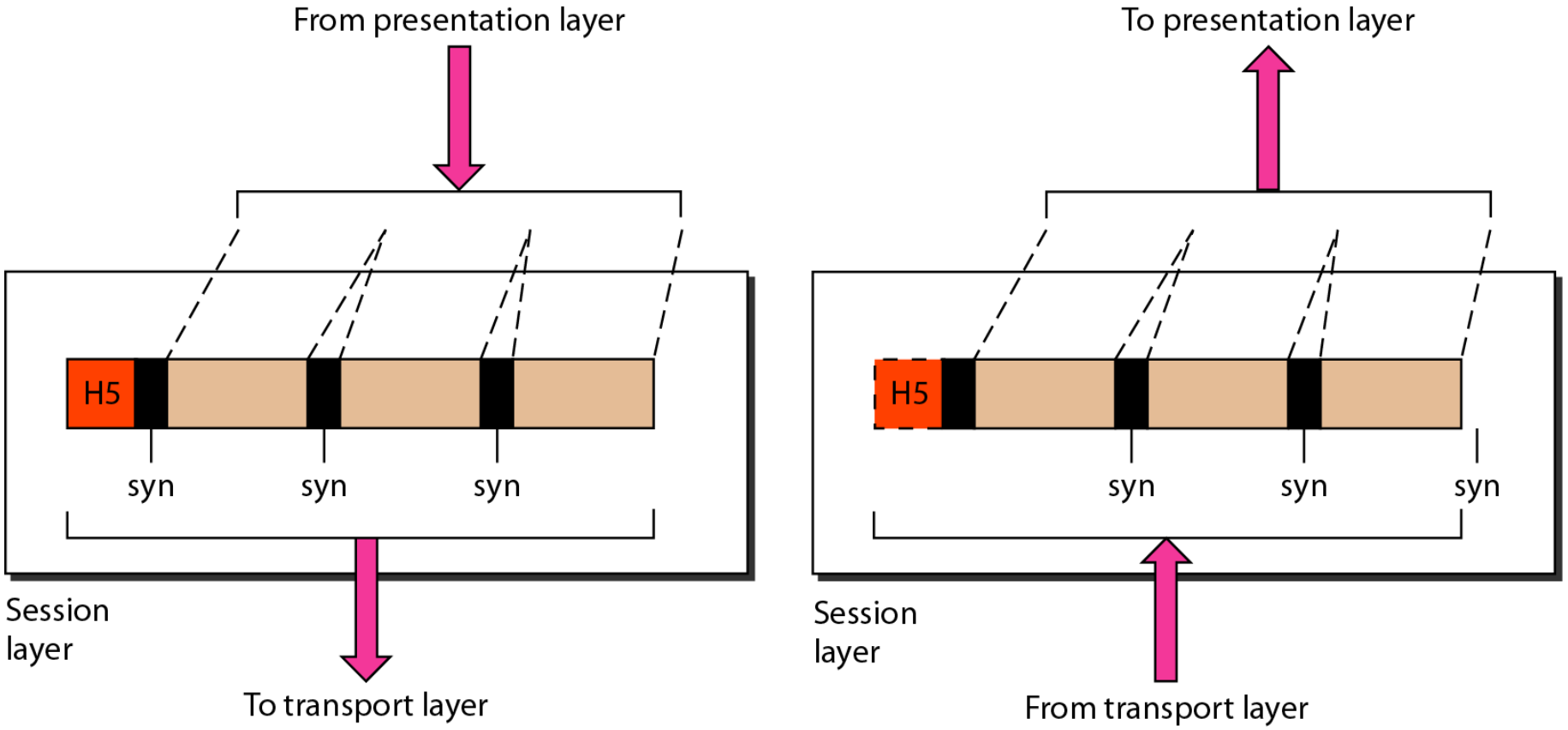


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The session layer is responsible for dialog control and synchronization.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Specific responsibilities of Session Layer includes the following:

1. Dialog Control
2. Synchronization: Checkpoints

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Presentation Layer

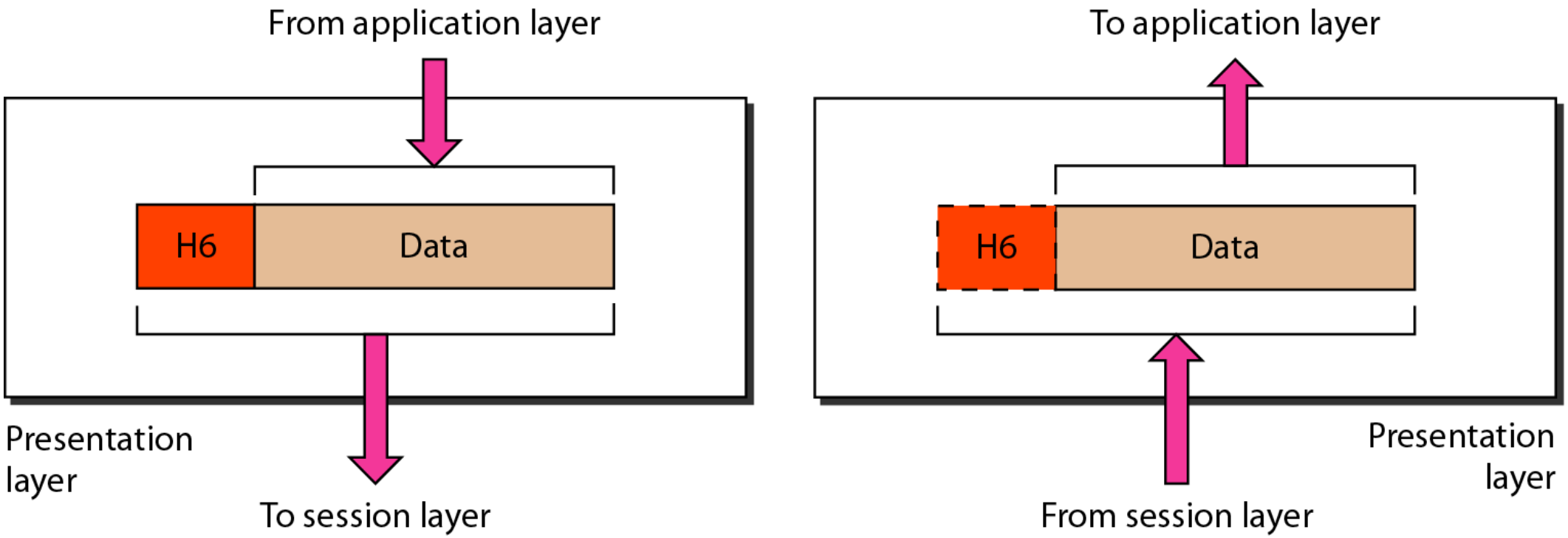


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The presentation layer is responsible for translation, compression, and encryption.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Specific responsibilities of Presentation Layer includes the following:

1. Translation
2. Encryption
3. Compression

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Application Layer

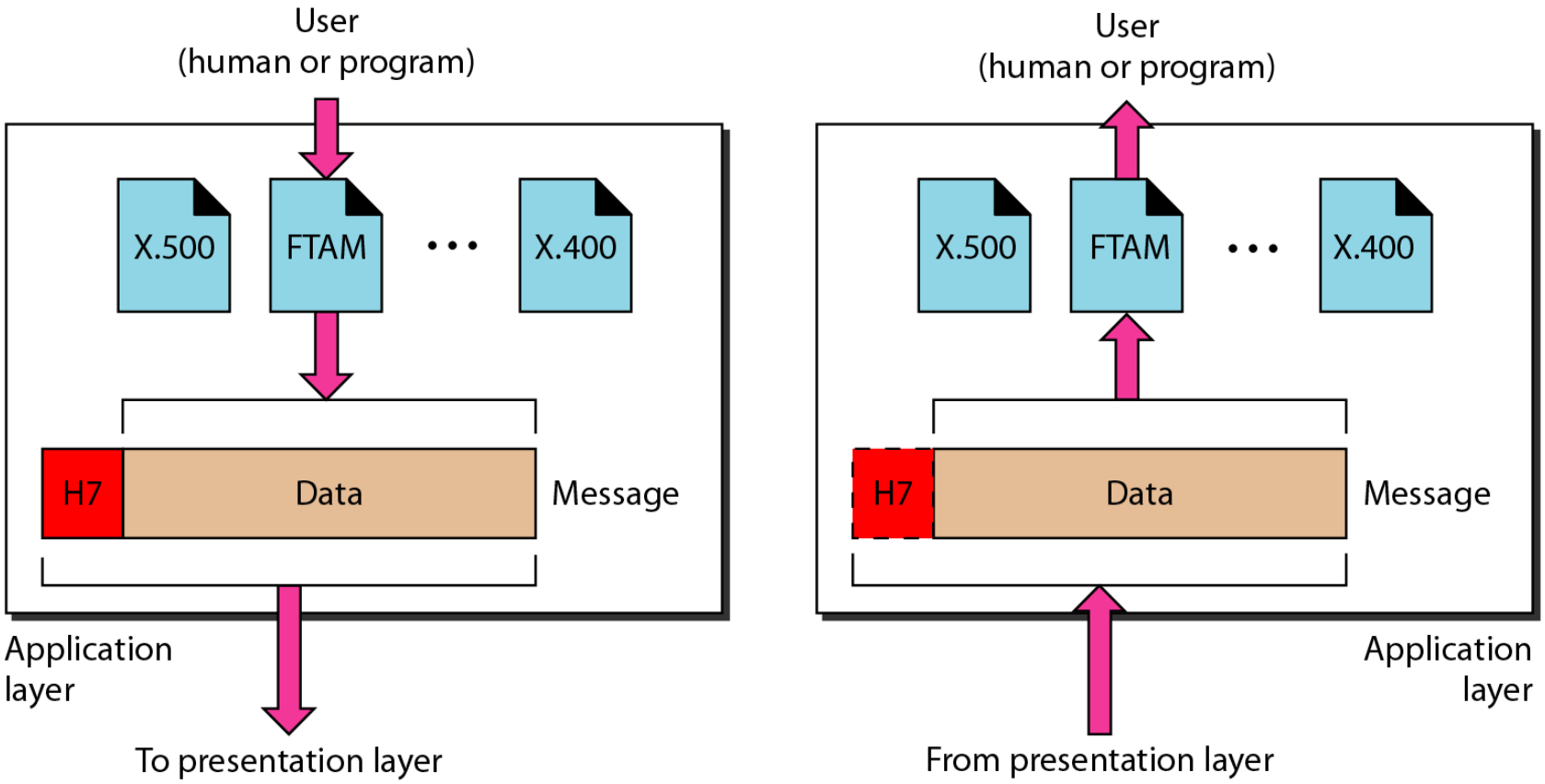


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Note

The application layer is responsible for providing services to the user.

Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Specific responsibilities of Presentation Layer includes the following:

1. Network virtual terminal
2. File transfer, access and management
3. Mail services
4. Directory services

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Layers Summery

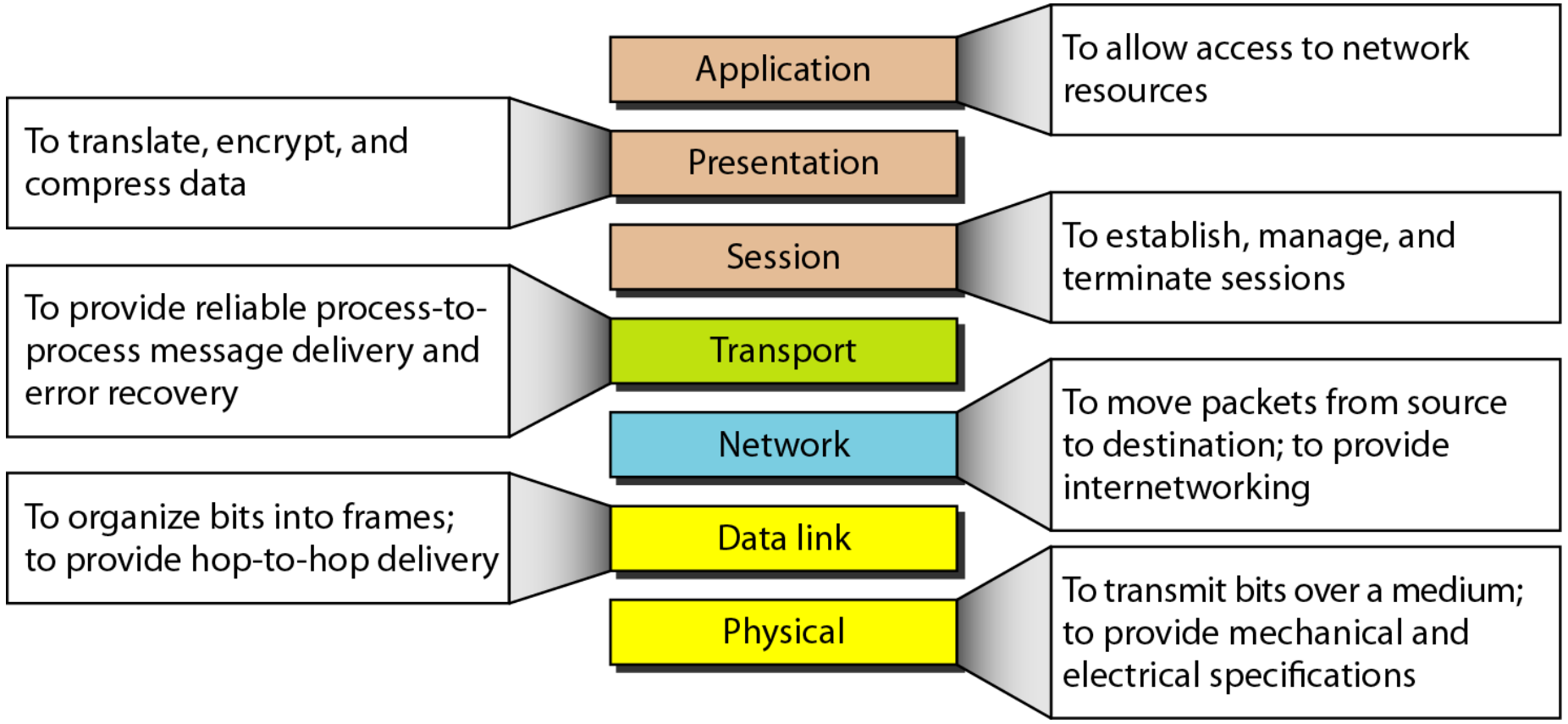


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

TCP/IP Protocol Suite

PROTOCOL LAYERING

TCP/IP PROTOCOL SUITE

- TCP/IP is a protocol suite used in the Internet today.
- A set of protocols organized in different layers.

- It is a hierarchical protocol made up of interactive modules,
- each module provides a specific functionality.

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TCP/IP PROTOCOL SUITE

- TCP/IP TCP/IP protocol suite was defined as four software layers built upon the hardware.

- *The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application.*

- *However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.*

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TCP/IP PROTOCOL SUITE LAYERS

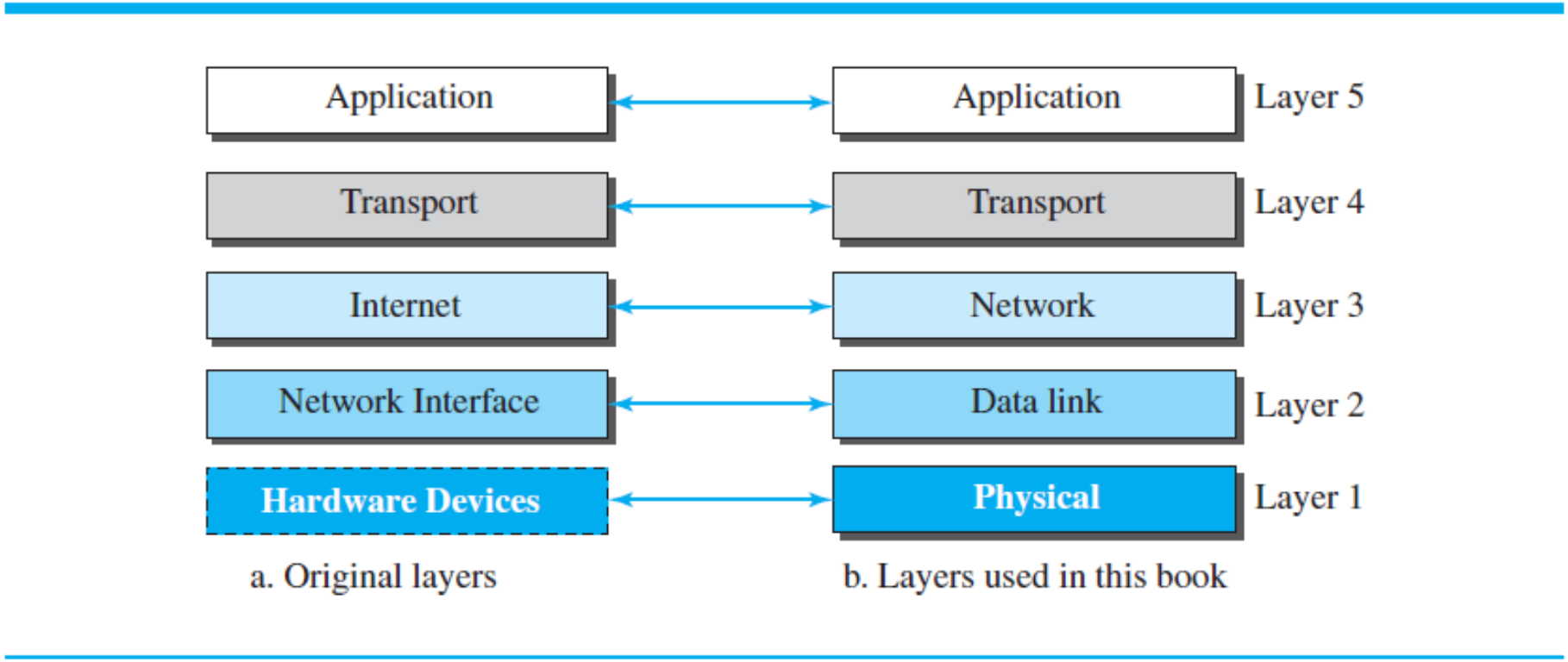


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

TCP/IP and OSI model

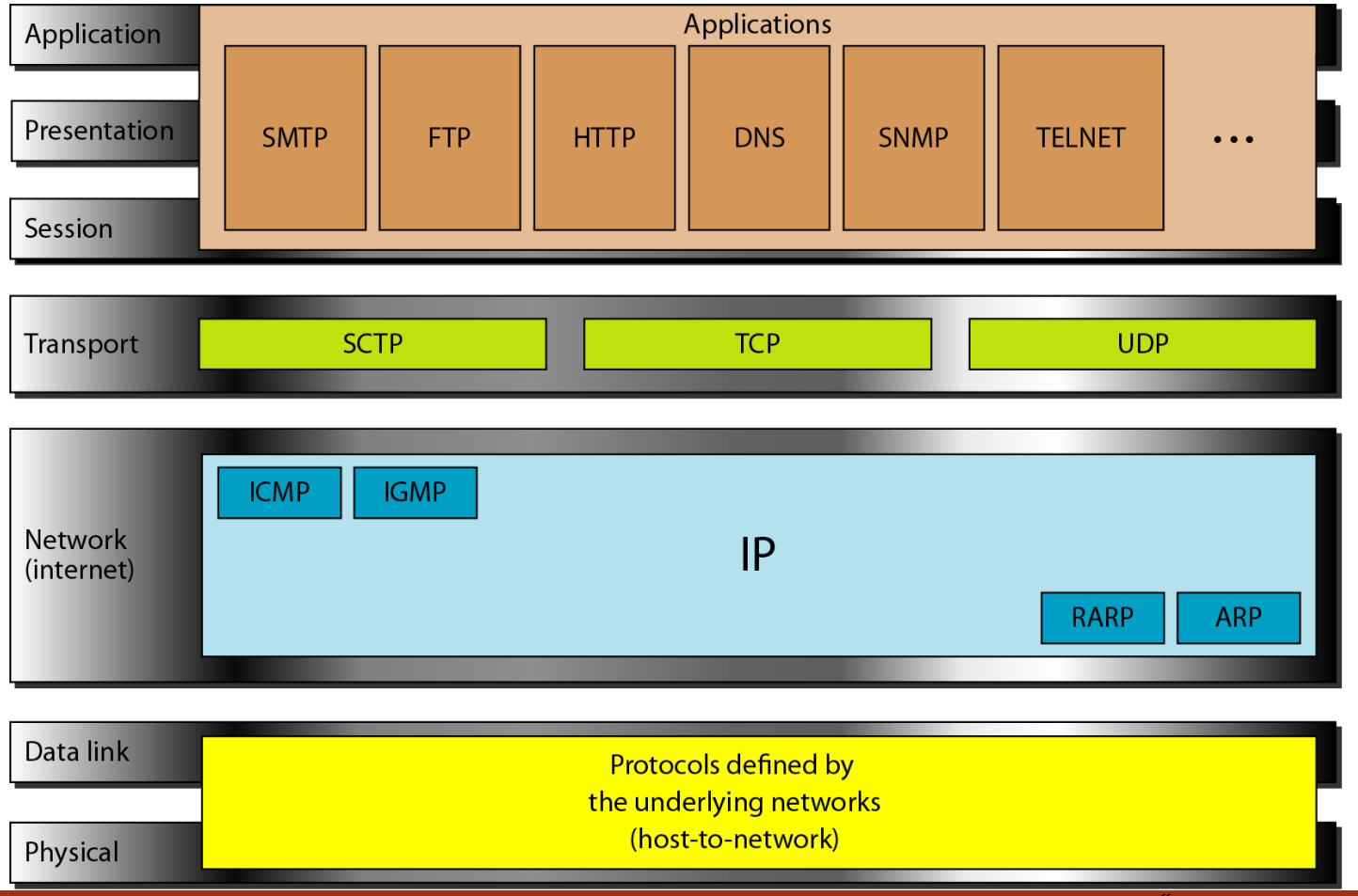


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Logical Connection

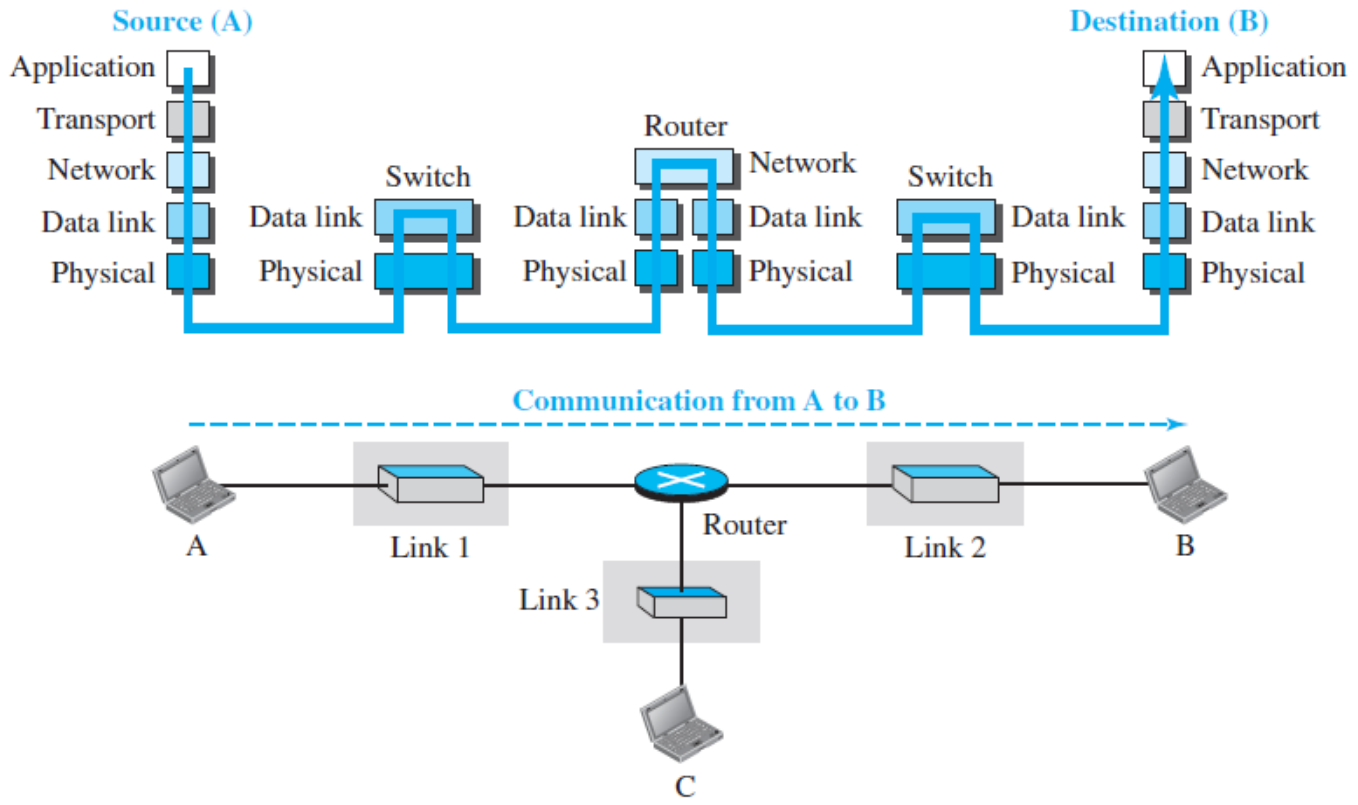


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

DESCRIPTION OF LAYERS (TCP/IP)

- The duty of the application, transport, and network layers is end-to-end.
- Domain of duty of the top three layers is the internet.
- In the top three layers, the data unit (packets) should not be changed by any router or link-layer switch.
- The duty of the data-link and physical layers is hop-to-hop.
- domain of duty of the two lower layers is the link.
- In the bottom two layers, the packet created by the host is changed only by the routers, not by the link-layer switches.

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Addresses in TCP/IP

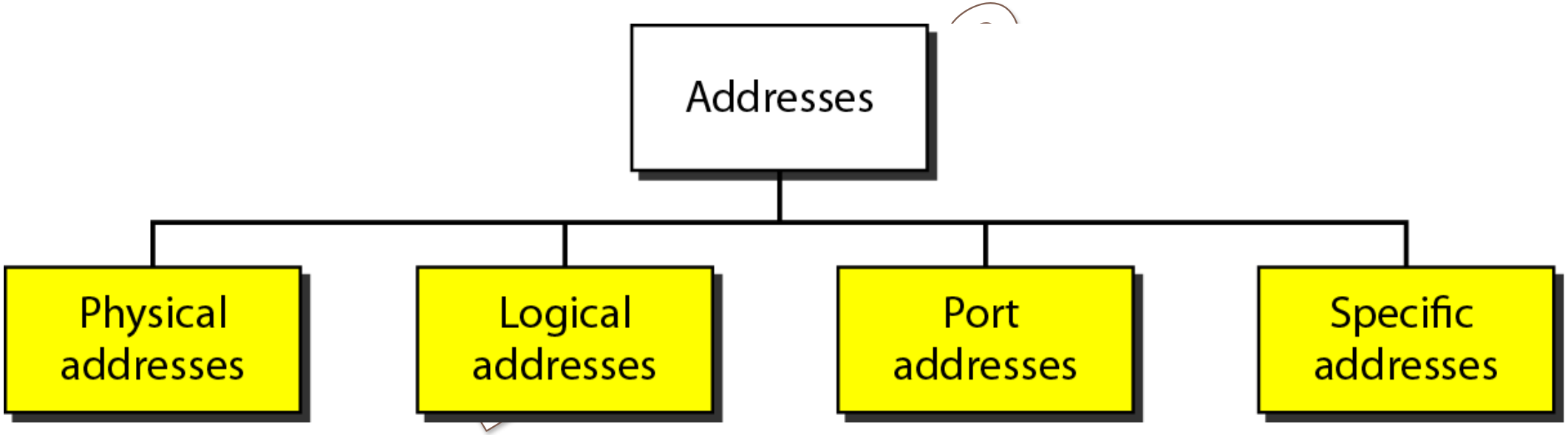


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

PHYSICAL AND DATA LINK LAYERS

- physical layer is responsible for carrying individual bits in a frame across the link.
- physical layer is the lowest level in the TCP/IP protocol suite,
- the communication between two devices at the physical layer is still a logical communication because there is another, hidden layer,
  - the transmission media, under the physical layer.
- we can think that the logical unit between two physical layers in two devices is a bit.
- There are several protocols that transform a bit to a signal.

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Physical Addresses

- The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN.
- It is included in the frame used by the data link layer.
- It is the lowest-level address.
- The physical addresses have authority over the network (LAN or WAN).
- The size and format of these addresses vary depending on the network

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Logical Addresses

- Logical addresses are necessary for universal communications that are independent of underlying physical networks.
- The logical addresses are designed for this purpose. A logical address in the Internet is currently a 32-bit address that can uniquely define a host connected to the Internet.
- No two publicly addressed and visible hosts on the Internet can have the same IP address.

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Port Addresses

- The IP address and the physical address are necessary for a quantity of data to travel from a source to the destination host.
- The end objective of Internet communication is a process communicating with another process.
- A port address in TCP/IP is 16 bits in length

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Relationship of layers and addresses in TCP/IP

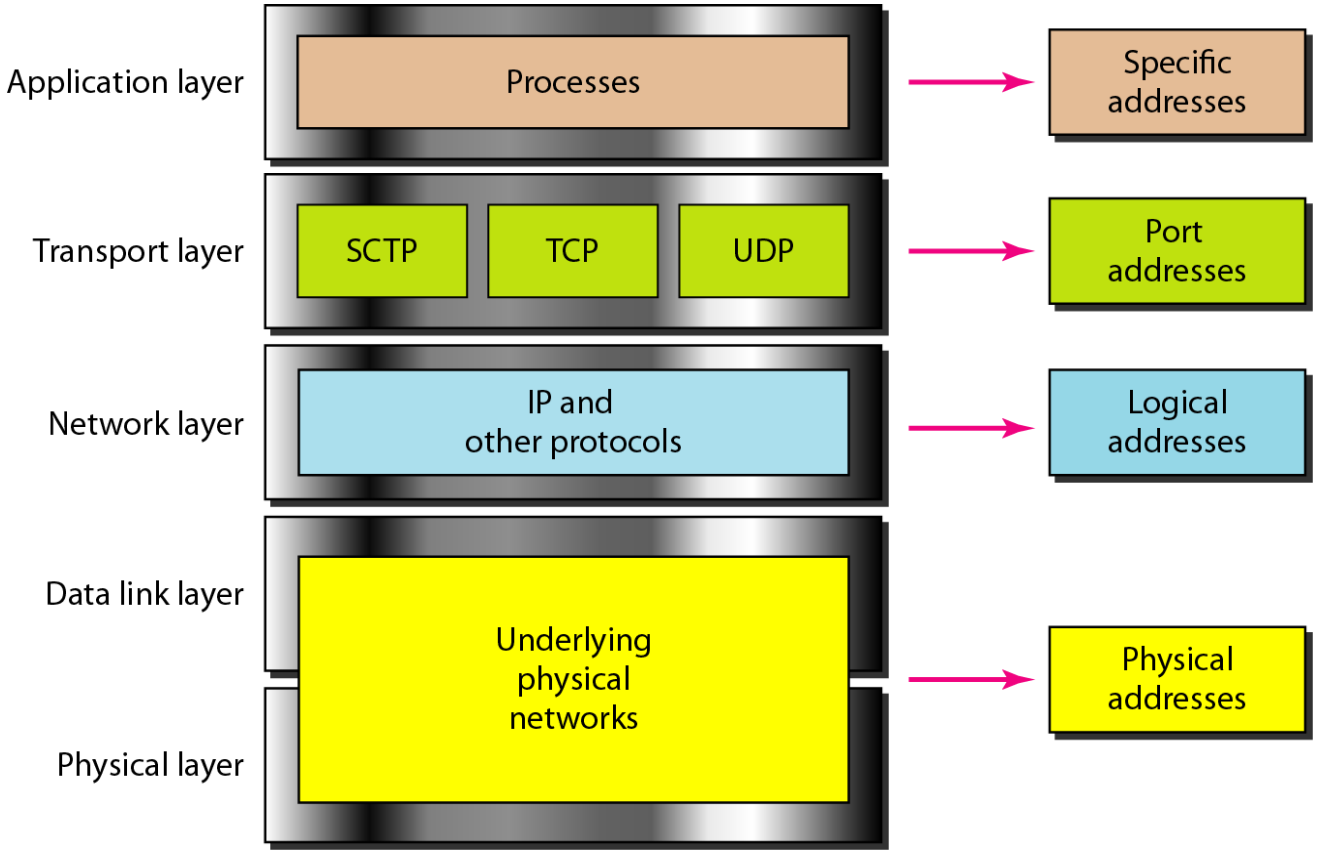


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

NETWORK LAYER

- At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

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Internetworking Protocol (IP)

- The Internetworking Protocol (IP) is the transmission mechanism used by the TCP/IP protocols.
- It is an unreliable and connectionless protocol–a best-effort delivery service.
- The term *best effort* means that IP provides no error checking or tracking.
- IP assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.
- IP transports data in packets called *datagrams,* each of which is transported separately.
- Datagrams can travel along different routes and can arrive out of sequence or be duplicated.

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Address Resolution Protocol

- The Address Resolution Protocol (ARP) is used to associate a logical address with a physical address.
- ARP is used to find the physical address of the node when its Internet address is known.

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Reverse Address Resolution Protocol

- The Reverse Address Resolution Protocol (RARP) allows a host to discover its Internet
- address when it knows only its physical address. It is used when a computer is connected
- to a network for the first time or when a diskless computer is booted.

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Refer Class Notes and Recommended Books for further Reading

Reference Books

- Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer networks* (5th ed.). Prentice Hall.
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- Kurose, J. F., & Ross, K. W. (2021). Computer networking: A top-down approach (8th ed.). Pearson.
- Peterson, L. L., & Davie, B. S. (2021). Computer networks: A systems approach (6th ed.). Morgan Kaufmann.
- Miller, M. A. (1992). Introduction to Digital and Data Communications. United States: West Publishing Company.
- Haykin, Simon, 3Rd Ed. (2008). Communication Systems India: Wiley India Pvt. Limited.
- Frenzel. (2016). Principles of Electronic Communication Systems. United Kingdom: McGraw-Hill Education.

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Thank You

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